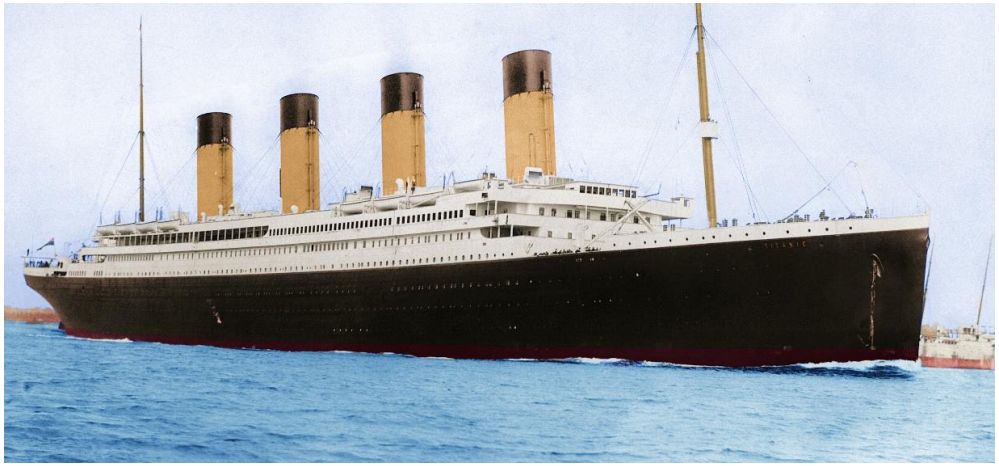
Practical activity 1



Dataset: titanic.csv
Python file: cardiacRiskLIME
Jupiter file: battacharya_LIME.ipynb

Applied Machine Learning Explainability Techniques
Aditya Bhattacharya
https://github.com/PacktPublishing/Applied-Machine-Learning-Explainability-Techniques/

- **Explain why a Titanic Passenger survived or not !**





- TITANIC
  - Predicting the survival of passengers on the Titanic
    - **Target** ={ death, survival }
  - Inputs/features

Variable	Description	Details
survival	Survival	0 = No; 1 = Yes
pclass	Passenger Class	1 = 1st; 2 = 2nd; 3 = 3rd
name	First and Last Name	
sex	Sex	
age	Age	
sibsp	Number of Siblings/Spouses Aboard	
parch	Number of Parents/Children Aboard	
ticket	Ticket Number	
fare	Passenger Fare	
cabin	Cabin	
embarked	Port of Embarkation	C = Cherbourg; Q = Queenstown; S = Southampton

- TITANIC
  - After pre-processing

Survived	Pclass	Sex	Age	SibSp	Parch	Fare	Embarked_0	Embarked_C	Embarked_Q	Embarked_S
0	3	1	22.0	1	0	7.2500	0	0	0	1
1	1	0	38.0	1	0	71.2833	0	1	0	0
1	3	0	26.0	0	0	7.9250	0	0	0	1
1	1	0	35.0	1	0	53.1000	0	0	0	1
0	3	1	35.0	0	0	8.0500	0	0	0	1

- **Why will this passenger survive (or not) ?**

1 · Compare different passengers

- Explain 5 test passengers, then 20.
- Record for each one:

Passenger	P(Survived)	Top 3 conditions (weights)	Intercept	Local prediction

- Which conditions contribute most to each prediction?
- Does a feature help some passengers and hurt others?
- With 20 passengers: which features appear most often?

2 · Check fidelity

- Does the explanation reproduce the model? For each passenger:

Passenger	P(Survived)	Local R^2 (explanation.score)	|P – local_pred|

- Which passengers are **poorly explained**?
- Is fidelity lower when the probability is near 0.5, or near 0 and 1?
- Does LIME explain the model worse for passengers where the model is undecided, or for those where it is very confident?
- Would you trust the explanation of the worst-explained passenger?

3 · Check stability

- Same passenger, **10 random seeds**, **three sample sizes:**

num_samples	Features selected (how often)	Mean weight	Range of weights
500			
1000			
5000			

- Are the **same features** selected every time?
- Do the **weights or signs** change?
- Does a larger **num_samples** make the explanation more stable? **At what cost in time?**

4 · Change the representation

- Explain one passenger with **discretize_continuous = True and False**.

- For insntance, how does the explanation of **Age** and **Fare** change?
- What does each number mean: effect of an interval, or slope per standard deviation?
- With False, does the sign of the bar match the effect on this passenger?

- **remember:** effect = coefficient × standardised value;

5 · Explore SP-LIME

- **Run SP-LIME with num_exps_desired = 3 and 5.**

- Which passengers are selected, and **how do they differ** from each other?
- Do the selected explanations cover all the important features?
- Run it again with another seed. Are the **same passengers** selected?
- Are the selected passengers **representative** of the dataset? Why or why not?

6 · Explore further

- Analyse any aspect you consider important.

7. Discussion and deliverable

Discuss

- Why can LIME give **different explanations for the same passenger**, even when the model's prediction does not change?
- What would you **check before showing** a LIME explanation to someone who must make a decision based on it?

Deliverable Explainability 1 (LIME)

- A report with your tables and plots, and some lines answering each question.